

Hardik Nandaniya

+91 6353691500 • hardiknandaniya811@gmail.com • [@LinkedIn](#) • [Github](#) • [Website](#)

SUMMARY

MSc Data Science student with 2+ years of experience in backend engineering, specializing in building scalable systems, real-time data pipelines, and high-performance APIs. Strong foundation in applied machine learning with hands-on experience in time-series forecasting, predictive modeling, and data-driven decision systems. Skilled in working with large-scale datasets, feature engineering, and model evaluation using both statistical and machine learning approaches. Experienced in integrating analytical solutions with production-grade backend systems, ensuring scalability, reliability, and performance. Passionate about designing and deploying end-to-end machine learning solutions that solve real-world problems and drive operational efficiency.

SKILLS

Data Science & Machine Learning

- Python, Pandas, NumPy, Matplotlib, Scikit-learn, PyTorch.
- Time Series Forecasting, Feature Engineering, Model Evaluation.
- ARIMA, Random Forest, XGBoost, LightGBM.

Backend & Systems

- Node.js, PHP, Javascript, Express.js.
- MongoDB, MySQL, Redis.
- REST APIs, System Design, Performance Optimization.

Tools & Technologies

- Jupyter Notebook, Tableau, Excel.
- Docker, Postman, JMeter.
- MQTT, AWS S3.

PROFESSIONAL EXPERIENCE

TECH INTEGRITY SERVICES.

Backend Developer

Rajkot, Guj

Jan 2023 - Oct 2024

- Designed and scaled backend systems handling **2K+ daily transactions and geospatial data** using Node.js, MongoDB, and Redis.
- Improved API response time by 20% through query optimization, caching strategies, indexing, and pagination.
- Engineered real-time data pipelines using Redis and MQTT processing **thousands of events per minute**.
- Designed and implemented a KPI reporting dashboard using MongoDB aggregation pipelines to analyze 15K+ order records, tracking order lifecycle metrics (completed vs cancelled orders), area-wise distribution, and time-based trends, enabling data-driven improvements in routing and scheduling.
- Implemented geospatial indexing for location-based analytics and delivery optimization.
- Conducted load testing using JMeter and resolved performance bottlenecks for high-traffic systems.
- Integrated AWS S3 and Twilio for notifications and media handling.

CRAFTBOX TECHNOLOGY.

Jr. Tech & Sales Representative

Rajkot, Guj

Feb 2022 - July 2022

- Developed web applications using PHP, HTML, CSS and JavaScript.
- Analyzed customer requirements to optimize software features and improve satisfaction.

PERSONAL PROJECTS

ZONE-LEVEL DEMAND FORECASTING USING ARIMA, XGBOOST & ENSEMBLE MODELS | [GITHUB](#).

- Developed time-series forecasting models on multi-zone food delivery dataset (20K+ records) using ARIMA, Random Forest, XGBoost, and LightGBM.
- Built ensemble model achieving $R^2 = 0.821$ and MAE = 5.74.
- Engineered 30+ features including lag variables, rolling statistics, and temporal features.
- Evaluated models using MAE, RMSE, MAPE, sMAPE, and MASE.
- Performed feature importance analysis to identify key demand drivers.
- Designed a data-driven system to support driver allocation and operational efficiency.

AurumAI – AI-Powered Gold Jewellery Design Identification System | [GITHUB](#).

- Developed an image retrieval system for gold jewellery design recognition using **ResNet50 embeddings(PyTorch)** and **FAISS cosine similarity search**.
- Built a modular **FastAPI backend** for image upload, embedding generation, vector search, and metadata mapping.
- Implemented deep feature extraction and normalized 2048-dimensional embeddings for similarity-based jewellery matching.
- Designed a scalable vector search pipeline using **FAISS Inner Product indexing** for nearest-neighbor retrieval.
- Created a web-based UI for uploading jewellery images and displaying top-K matching designs with confidence scores.
- Applied image preprocessing, embedding normalization, and cosine similarity techniques to improve retrieval accuracy.

E-COMMERCE BACKEND SYSTEM | [GITHUB](#).

- Built scalable backend using Node.js and TypeScript with clean architecture supporting multiple concurrent users and transactions.
- Architected modules for products, carts, orders, and authentication.
- Optimized system performance and maintainability.

EDUCATION

MARWADI UNIVERSITY

Master of Science in Data Science

Rajkot, Guj
Pursuing

SILVER OAK UNIVERSITY

Bachelor of Vocation in Information Technology With 8.0 CGPA

Ahmedabad, Guj
2021-2024

JAWAHAR NAVODAYA VIDHYALAYA

HSC - Science

Ahmedabad, Guj
2020-2021

ADDITIONAL INFORMATION

- **Languages:** English (professional), Hindi (fluent), Gujarati (native).
- **Certifications:**
 - Advanced Statistics and Data Mining for Data Science — Infosys Springboard (2026).
 - Data Mining with Python: Classification & Regression — Infosys Springboard (2026).
 - Machine Learning using Python — Infosys Springboard (2026).
 - Hands-on Machine Learning with Scikit-learn & TensorFlow — Infosys Springboard (2026).